

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Explains why $\angle EFQ = A$ Must be a fully correct explanation with reasons which may include: Vertically opposite angles and right angle implies similar triangles.	AO2.4	E1	$\angle OQR = \angle FQE$ vertically opposite angles $\angle ORQ = \angle FEQ = 90^\circ$ So $\angle EFQ = A$
	Deduces $\frac{PF}{EF} = \cos(A)$ AND $\frac{EF}{OF} = \sin(B)$ Must have at least stated or implied that $\angle EFQ = A$ through similarity	AO2.2a	R1	Since $\angle EFQ = A$ $\frac{PF}{EF} = \cos(A)$ And $\frac{EF}{OF} = \sin(B)$ in triangle OEF
(b)	Completes proof	AO2.2a	B1	$\frac{DE}{OE} \times \frac{OE}{OF} + \frac{PF}{EF} \times \frac{EF}{OF}$ $= \sin A \cos B + \cos A \sin B$
(c)	Explains that the proof is based on right angled triangles which limits A and B to acute angles	AO2.3	E1	Since the proof is based on the diagram which uses right-angled triangles it is assumed that A and B are acute. Therefore, the proof only holds for acute angles.
(d)	Substitutes $-B$ into identity for $\sin(A + B)$ to give $\sin(A - B)$	AO2.1	R1	$\sin(A - B) = \sin A \cos(-B) + \cos A \sin(-B)$
	Recalls at least one of the identities $\sin(-B) = -\sin(B)$ $\cos(-B) = \cos(B)$ Must be explicitly stated	AO1.2	B1	$\sin(-B) = -\sin(B)$ $\cos(-B) = \cos(B)$
	Deduces correct identity with no errors. This must be clearly deduced from a correct argument and not simply stated.	AO2.2a	R1	Hence $\sin(A - B) = \sin A \cos B - \cos A \sin B$

Total 7 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Compares with $R \cos(x \pm \alpha)$ or $R \sin(x \pm \alpha)$	AO3.1a	M1	$\sqrt{3} \sin x - 3 \cos x = R \sin(x - \alpha)$ $= R \sin x \cos \alpha - R \cos x \sin \alpha$ $R \cos \alpha = \sqrt{3}$ $R \sin \alpha = 3$ $R = \sqrt{12} = 2\sqrt{3}$ $\tan \alpha = \sqrt{3}$ $\alpha = \frac{\pi}{3}$ $y = 2\sqrt{3} \sin\left(x - \frac{\pi}{3}\right) + 4$ Translation $\begin{pmatrix} \frac{\pi}{3} \\ 0 \end{pmatrix}$ Stretch in the y -direction scale factor $2\sqrt{3}$ Translation $\begin{pmatrix} 0 \\ 4 \end{pmatrix}$
	Obtains two correct equations for R and α for example $R \cos \alpha = \sqrt{3}$ $R \sin \alpha = 3$ Must be explicitly seen	AO3.1a	M1	
	Obtains correct R Condone AWR 3.46 PI by description of stretch	AO1.1b	B1	
	Obtains correct α in radians or degrees PI by description of translation	AO1.1b	B1	
	Interprets their values of R and α to form an equation of the form $y = R \sin(x \pm \alpha) + 4$ or $y = R \cos(x \pm \alpha) + 4$	AO3.2a	B1F	
	Interprets 'their' equation to identify a transformation	AO3.2a	E1F	
	Identifies all required transformations in a correct order CAO	AO3.2a	A1	
(b)	(i) Deduces the least value occurs when their $\sin\left(x - \frac{\pi}{3}\right) = 1$ Using 'their' values of R and α PI by sight of $\frac{1}{2\sqrt{3} + 4}$	AO2.2a	M1	$\frac{1}{\sqrt{3} \sin x - 3 \cos x + 4} = \frac{1}{2\sqrt{3} \sin\left(x - \frac{\pi}{3}\right) + 4}$ Least value when $\sin\left(x - \frac{\pi}{3}\right) = 1$ $\therefore$ least value is given by $\frac{1}{2\sqrt{3} + 4} = \frac{2 - \sqrt{3}}{2}$
	Completes rigorous argument to obtain	AO2.1	R1	

$\frac{1}{2\sqrt{3}+4}$ and then the given answer			
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(ii)	Deduces the greatest value using 'their' values of R and α $\text{ACF } \frac{1}{-2\sqrt{3}+4} = \frac{2+\sqrt{3}}{2}$	AO2.2a	B1F	Greatest value = $\frac{2+\sqrt{3}}{2}$
Total 10 marks				

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Recalls a correct trig identity, which could lead to a correct answer	AO1.2	B1	$\begin{aligned} &(\text{LHS} \equiv) \\ &\frac{\sin 2x}{1 + \tan^2 x} \\ &= \frac{2 \sin x \cos x}{1 + \tan^2 x} \\ &= \frac{2 \sin x \cos x}{\sec^2 x} \\ &\equiv 2 \sin x \cos x \cos^2 x \\ &\equiv 2 \sin x \cos^3 x \\ &(\equiv \text{RHS}) \end{aligned}$
	Demonstrates a strategy for proving the identity, eg by converting all the terms on the LHS to cos and sin.	AO3.1a	M1	
	Concludes a rigorous mathematical argument to prove given identity AG	AO2.1	R1	
(b)	Uses identity to write integrand in the form $a \sin 2\theta \cos^3 2\theta$	AO1.1a	M1	$\begin{aligned} &\int \frac{4 \sin 4\theta}{1 + \tan^2 2\theta} d\theta = \int 8 \sin 2\theta \cos^3 2\theta d\theta \\ &\text{Let } u = \cos 2\theta \\ &\text{then } \frac{du}{d\theta} = -2 \sin 2\theta \Rightarrow \sin 2\theta = -\frac{1}{2} \frac{du}{d\theta} \\ &I = -4 \int u^3 \frac{du}{d\theta} d\theta \\ &= -4 \int u^3 du \\ &= -u^4 + c \\ &= -\cos^4 2\theta + c \end{aligned}$
	Correctly writes integrand as $8 \sin 2\theta \cos^3 2\theta$	AO1.1b	A1	
	Selects an appropriate method for integrating, e.g. substitution $u = \cos 2\theta$, or by inspection PI by sight of $\cos^4 2\theta$	AO3.1a	M1	
	Obtains $k \int u^3 du$ correctly PI by solution in form $k \cos^4 2\theta$, if by inspection	AO1.1a	M1	
	Obtains $-u^4$ or $-\cos^4 2\theta$	AO1.1b	A1F	

OE Only FT value of a			
Completes rigorous argument to obtain $-\cos^4 2\theta + c$ OE	AO2.1	R1	
Total 9 marks			

Q4.

Marking Instructions	AO	Marks	Typical Solution
Translates $f\left(\frac{\pi}{6}\right)$ into $\lim_{h \rightarrow 0} \frac{\sin\left(\frac{\pi}{6} + h\right) - \sin\left(\frac{\pi}{6}\right)}{h}$	AO1.1a	M1	$f\left(\frac{\pi}{6}\right) = \lim_{h \rightarrow 0} \left[\frac{\sin\left(\frac{\pi}{6} + h\right) - \sin\left(\frac{\pi}{6}\right)}{h} \right]$
Uses $\sin(A+B)$ identity to replace $\sin\left(\frac{\pi}{6} + h\right)$, to commence argument (at least two lines of argument seen)	AO2.1	M1	$= \lim_{h \rightarrow 0} \left[\frac{\sin\frac{\pi}{6}\cos h + \cos\frac{\pi}{6}\sin h - \sin\left(\frac{\pi}{6}\right)}{h} \right]$
Obtains correct two term expression involving $\cos h$ and $\sin h$	AO2.5	M1	$= \lim_{h \rightarrow 0} \left[\frac{\frac{1}{2}\cos h + \frac{\sqrt{3}}{2}\sin h - \frac{1}{2}}{h} \right]$
Deduce what happens as $h \rightarrow 0$, for one part of 'their' expression using the limit of $\frac{\sin h}{h}$ OR by using small angle approximations	AO2.2a	R1	$= \lim_{h \rightarrow 0} \left[\frac{1}{2} \left(\frac{\cos h - 1}{h} \right) + \frac{\sqrt{3}}{2} \frac{\sin h}{h} \right]$
Deduce what happens as $h \rightarrow 0$, for the second part of 'their' expression using the limit of $(\cos h - 1)h$ OR by using small angle approximations	AO2.2a	R1	$= \lim_{h \rightarrow 0} \left[\frac{-2\sin^2\left(\frac{h}{2}\right)}{2h} \right] + \frac{\sqrt{3}}{2} \frac{\sin h}{h}$
Completes a rigorous argument leading to the	AO2.1	R1	$= \lim_{h \rightarrow 0} \left[\frac{-\sin\left(\frac{h}{2}\right)}{2} \right] \lim_{h \rightarrow 0} \left[\frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)} \right] + \frac{\sqrt{3}}{2} \lim_{h \rightarrow 0} \frac{\sin h}{h}$
			$= 0 \times 1 + \frac{\sqrt{3}}{2} \times 1$
			$= \frac{\sqrt{3}}{2}$

correct exact value, with all the steps in the method clearly shown.			
Total 6 marks			

Q5.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Compares with $R \cos(\theta \pm \alpha)$ or $R \sin(\theta \pm \alpha)$	AO3.1a	M1	$R \cos(\theta - \alpha)$ $\equiv R \cos \alpha \cos \theta + R \sin \theta \sin \alpha$ $\therefore R \cos \alpha = 3$ and $R \sin \alpha = 3$ $R = \sqrt{18}$ $\alpha = \frac{\pi}{4}$ $\therefore 3 \cos \theta + 3 \sin \theta \equiv \sqrt{18} \cos\left(\theta - \frac{\pi}{4}\right)$ Which is a stretch in the y -direction scale factor $\sqrt{18}$ $\left(\frac{\pi}{4}\right)$ and a translation $\left(0\right)$
	Identifies version which will allow them to solve the problem	AO3.1a	A1	
	Obtains correct R	AO1.1b	A1	
	Obtains correct α	AO1.1b	A1	
	Interprets 'their' equation to identify first transformation	AO3.2a	E1	
	Interprets 'their' equation to identify second transformation	AO3.2a	E1	
(b)	Constructs a rigorous mathematical argument, to find either the least or greatest value Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips (no FT for this mark)	AO2.1	R1	$4 + (3 \cos \theta + 3 \sin \theta)^2$ $4 + \left(\sqrt{18} \cos\left(\theta - \frac{\pi}{4}\right)\right)^2$ Least value occurs when $\cos^2\left(\theta - \frac{\pi}{4}\right) = 0$ $\therefore \text{least value} = 4$ Greatest value occurs when $\cos^2\left(\theta - \frac{\pi}{4}\right) = 1$ Greatest value $= 4 + 18 = 22$
	Deduces the least value Using 'their' values of R and α	AO2.2a	A1F	
	Deduces the greatest value Using 'their' values of R and α	AO2.2a	A1F	

Q6.

(a) (i) $R = \sqrt{13}$
Accept 3.6 or better

B1

$$\tan \alpha = \frac{2}{3}$$

OE

M1

$$\alpha = 33.7^\circ$$

A1

3

(ii) minimum value = $-\sqrt{13}$
Accept - 3.6 or better; ft R

B1ft

when $x - \alpha = \cos^{-1}(-1)$
NMS 0 / 2

M1

$$x = 213.7^\circ$$

Calculus used 0 / 2

A1

3

(b) (i) $\text{LHS} = \frac{\cos x}{\sin x} - 2 \sin x \cos x$
Express $\cot x - \sin 2x$ in terms of $\sin x$ and $\cos x$; ACF

M1

$$= \frac{\cos x}{\sin x} (1 - 2 \sin^2 x)$$

Factor out $\frac{\cos x}{\sin x}$ and $1 - 2 \sin^2 x$

m1

$$= \cot x \cos 2x$$

All correct

A1

3

Alternatives:

$$\text{RHS} = \cot x \cos 2x$$

$$= \frac{\cos x}{\sin x} (1 - 2\sin^2 x)$$

Express $\cot x \cos 2x$ in terms of $\cos x$ and $\sin x$,
 $\cos 2x$ ACF

(M1)

$$= \frac{\cos x}{\sin x} - 2\sin x \cos x$$

$\cos 2x = 1 - 2\sin^2 x$ and multiply out and simplify.

(m1)

$$= \cot x - \sin 2x$$

All correct.

(A1)

(3)

$$\cot x(1 - \cos 2x) - \sin 2x = 0$$

$$\frac{\cos x}{\sin x} (1 - (1 - 2\sin^2 x)) - 2\sin x \cos x = 0$$

Rearrange to expression = 0 and factor out $\cot x$;

Express $\cot x$, $\cos 2x$ and $\sin 2x$ in terms of $\sin x$ and $\cos x$, ACF

(M1)

$$\frac{\cos x}{\sin x} (2\sin^2 x) - 2\sin x \cos x = 0$$

$\cos 2x = 1 - 2\sin^2 x$ used

(m1)

$$2\sin x \cos x - 2\sin x \cos x = 0$$

Simplified, with all correct

(A1)

(3)

(ii) $\cot x - \sin 2x = 0$

$$\cot x \cos 2x = 0$$

$$\cot x = 0 \quad \text{or} \quad \cos 2x = 0$$

Both equations correct

M1

$$2x = 90^\circ \quad (270^\circ)$$

Condone missing 270°

m1

$$x = 90^\circ, 45^\circ, 135^\circ$$

All correct

A1

3

Alternatives:

$$\cot x - \sin 2x = \frac{\cos x}{\sin x} - 2\sin x \cos x = 0$$

$$\cos x \left(\frac{1}{\sin x} - 2\sin x \right) = 0$$

$$\cos x = 0 \quad \text{or} \quad 1 - 2\sin^2 x = 0$$

Both equations

(M1)

$$\sin x = (\pm) \frac{1}{\sqrt{2}}$$

(m1)

$$x = 90^\circ, 45^\circ, 135^\circ$$

(A1)

(3)

[12]

Q7.

$$(a) \quad (i) \quad h^2 = 2^2 + \sqrt{5}^2 = 9 \Rightarrow h = 3 \Rightarrow \sin \alpha = \frac{2}{3}$$

Pythagoras used or all of 2, $\sqrt{5}$, 3 seen correctly on triangle

B1

$$\cos \alpha = \frac{\sqrt{5}}{3}$$

AG $\frac{\sqrt{5}}{3}$ or $\sqrt{\frac{5}{9}}$ or $\frac{5}{3\sqrt{5}}$ seen

B1

2

Alternative:

$$\operatorname{cosec}^2 \alpha = 1 + \cot^2 \alpha = 1 + \frac{5}{4} = \frac{9}{4}$$

$$\operatorname{cosec} \alpha = \frac{3}{2} \quad \sin \alpha = \frac{2}{3}$$

Must be positive

(B1)

$$\sec^2 \alpha = 1 + \tan^2 \alpha = 1 + \frac{4}{5} = \frac{9}{5}$$

$$\sec \alpha = \frac{3}{\sqrt{5}} \quad \cos \alpha = \frac{\sqrt{5}}{3}$$

Must be positive

(B1)

(2)

(ii) $\sin 2\alpha = 2 \sin \alpha \cos \alpha$

Correct formula seen or implied

M1

$$= \left(2 \times \frac{2}{3} \times \frac{\sqrt{5}}{3} \right) = \frac{4}{9} \sqrt{5}$$

Must see $\frac{\sqrt{5}}{3}$ here or in part (a)(i)

Accept $\frac{4}{3} \sqrt{\frac{5}{9}}$

A1

2

(b) $\cos \beta = \frac{2}{\sqrt{5}}$ or $\sin \alpha = \frac{1}{\sqrt{5}}$

Either correct. Accept $\sqrt{\frac{4}{5}}$, $\frac{\sqrt{5}}{5}$

B1

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

Correct formula seen or implied.

M1

$$= \frac{\sqrt{5}}{3} \times \frac{2}{\sqrt{5}} + \frac{2}{3} \times \frac{1}{\sqrt{5}}$$

All correct

A1

$$= \frac{2}{15} (5 + \sqrt{5})$$

k = 5 with previous A mark awarded

A1

4

[8]

Q8.

(a) $f\left(-\frac{3}{2}\right) = 4\left(-\frac{3}{2}\right)^3 - 11\left(-\frac{3}{2}\right) - 3$

x = -\frac{3}{2} substituted

M1

$$= -4 \times \frac{27}{8} + \frac{33}{2} - 3 = 0 \Rightarrow \text{factor}$$

Processing, = 0 and conclusion

A1

2

(b) $2x^2 - 3x - 1$

M1 for any two of a, b, c correct

M1A1

2

(c) (i) $2\cos 2\theta \sin \theta + 9\sin \theta + 3$
 $= 2(1 - 2\sin^2 \theta)\sin \theta + 9\sin \theta + 3$

cos 2θ expanded; ACF and substituted

M1

$$= 2\sin \theta - 4\sin^3 \theta + 9\sin \theta + 3$$

All in terms of sinθ or x and simplified to a cubic expression

m1

$$\sin \theta = x \Rightarrow 4x^3 - 11x - 3 = 0$$

Reverse signs and express in x correctly **AG**

A1

3

(ii) $2x^2 - 3x - 1 = 0 \Rightarrow x = \frac{3 \pm \sqrt{17}}{4}$

Use formula correctly to solve $ax^2 + bx + c = 0$ from part (b)

M1

$x = \frac{3 - \sqrt{17}}{4}$ or $-0.28\dots$

A1

$\theta = 196^\circ$ and 344°

Both required and no others in range; condone greater accuracy. Ignore solutions out of range.

A1

$x = \frac{3 + \sqrt{17}}{4}$ no solutions for $\sin \theta$

$x = -\frac{3}{2}$ no solutions for $\sin \theta$

Must have three correct roots and reject both other roots from cubic equation

E1

4

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[11]

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Q9.

(a) (i) $\tan \alpha = \frac{4}{3}$

Fraction required

Allow 1.333 (recurring)

B1

1

(ii) $1, 2, \sqrt{3}$ seen (from Pythagoras)
or

$4 = 1 + \cot^2 \beta$

Use $\operatorname{cosec}^2 \beta = 1 + \cot^2 \beta$

M1

$$\tan \beta = -\frac{1}{\sqrt{3}}$$

$$\text{SC B1 } \frac{1}{\sqrt{3}} \text{ or } \frac{\sqrt{3}}{3}$$

A1

$$\text{Special case B1 for } \frac{1}{\sqrt{3}} \text{ or } \frac{\sqrt{3}}{3}$$

2

$$(b) \quad \tan(\alpha + \beta) = \frac{\frac{4}{3} - \frac{1}{\sqrt{3}}}{1 - \frac{4}{3} \left(-\frac{1}{\sqrt{3}} \right)}$$

Use $\tan(\alpha + \beta)$ formula

M1

Remove fractions within fractions

$$\text{Correct manipulation to form } \frac{a+b\sqrt{3}}{c+d\sqrt{3}} \quad a \ b \ c \ d \text{ integers}$$

m1

$$= \frac{4\sqrt{3}-3}{3\sqrt{3}+4}$$

$$m = 4 \quad n = 3$$

or any multiple

EXAM PAPERS PRACTICE

3

Alternative $\tan(\alpha + \beta)$

$$\frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)} = \frac{\frac{4}{5} \times \left(-\frac{\sqrt{3}}{2} \right) + \frac{3}{5} \times \frac{1}{2}}{\frac{3}{5} \times \left(-\frac{\sqrt{3}}{2} \right) - \frac{4}{5} \times \frac{1}{2}}$$

Use formulae for $\sin(\alpha + \beta)$ and $\cos(\alpha + \beta)$

(M1)

Remove fractions within fractions

$$\text{Correct manipulation to form } \frac{a+b\sqrt{3}}{c+d\sqrt{3}} \quad a \ b \ c \ d \text{ integers}$$

(m1)

$$= \frac{-4\sqrt{3}+3}{-3\sqrt{3}-4} \left(= \frac{4\sqrt{3}-3}{3\sqrt{3}+4} \right)$$

$m = -4 \quad n = -3$
or any multiple

(A1)

(3)

M1 for substituting candidates values for $\tan \alpha$ and $\tan \beta$ into correct formula.
Completely correct or completely correct ft on $\tan \alpha$, $\tan \beta$.

Special case answer is $\frac{12+3\sqrt{3}}{9-4\sqrt{3}}$ or $\times \frac{a}{a}$ where a is integer or $\sqrt{3}$ for M1m1A0.

[6]

Q10.

(a) $16\left(\frac{3}{4}\right)^3 + 11\left(\frac{3}{4}\right) - 15$

Evaluate $f\left(\frac{3}{4}\right)$ not long division.

M1

$$= \frac{27}{4} + \frac{33}{4} - 15 = 0 \Rightarrow \text{factor}$$

Processing and conclusion.

A1

For A1; minimum processing seen; $16 \times \frac{27}{64} + 11 \times \frac{3}{4} - 15 = 0$; $15 - 15 = 0$
and no other working is A0 minimum conclusion = 0 hence factor reserved.

2

(b) $27\cos\theta(2\cos^2\theta - 1) +$

Use acf of $\cos 2\theta$ formula

B1

$$19\sin\theta(2\sin\theta\cos\theta) - 15 = 0$$

Use acf of $\sin 2\theta$ formula

B1

$$54\cos^3\theta - 27\cos\theta + 38(1 - \cos^2\theta)\cos\theta - 15 = 0$$

All in cosines.

M1

$$16\cos^3\theta + 11\cos\theta - 15 = 0$$

$$x = \cos\theta \Rightarrow 16x^3 + 11x - 15 = 0$$

Simplification and substitute $x = \cos\theta$ to obtain AG CSO.

A1

For M1 mark; $\cos 2\theta$ (eventually) in form $a \cos^2\theta + b$; $19\sin\theta \sin 2\theta$ in form $c \cos\theta \sin^2\theta$ and use $\sin^2\theta = 1 - \cos^2\theta$ to obtain $c \cos\theta (1 - \cos^2\theta)$

4

(c) $16x^3 + 11x - 15 = (4x - 3)(4x^2 + 3x + 5)$

Factorise $f(x)$

M1A1

$$b^2 - 4ac = 3^2 - 4 \times 4 \times 5 \quad (= -71)$$

Find discriminant of quadratic factor; or seen in formula

m1

$$b^2 - 4ac \ x^2 + 3x + 5 = 0)$$

Conclusion; CSO

$$\Rightarrow \text{(only) solution is } \cos\theta = \frac{3}{4}$$

Condone $\frac{3}{4}$ is (only) solution

A1

M1 $(4x - 3)(4x^2 + kx \pm 5)$ A1 fully correct

m1 candidate's values of a, b, c used in expression for $b^2 - 4ac$

or complete square to obtain
$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2}{4a^2} - \frac{c}{a}$$

A1 $b^2 - 4ac$ correct or
$$\left(x + \frac{3}{8}\right)^2 = \frac{9}{64} - \frac{5}{4} \quad \left(= -\frac{71}{64}\right)$$
 and stated

*to be negative so no solution or solutions are not real (imaginary)
Accept imaginary solutions from calculator if stated to be imaginary.*

Condone $\sqrt{-71}$ is negative, or similar, so no solution.

Conclusion $x = \frac{3}{4}$ is solution, or $\cos\theta = \frac{3}{4}$ is solution.

4

[10]

Q11.

(a) $R = \sqrt{10}$

Accept 3.2 or better. Can be earned in (b)

B1

$\tan \alpha = 3$

OE; M0 if $\tan \alpha = -3$ seen

M1

$\alpha = 71.6$ or better

$\alpha = 71.56505\dots$

A1

3

(b) $\sin(x \pm \alpha) = \frac{-2}{R}$

or their R and / or their α ; PI

M1

$x(= -39.2 + 71.6) = 32(.333)$

32 or better

Condone 32.4

A1

or

$x - 71.6 = 219.2$

*must see 219 and 72 or better PI by 291 or better as answer
Condone extra solutions*

m1

$x = 291$

*Condone 290.8 or better CSO Withhold final A1 if more
than two answers given within interval*

A1

4

[7]

Q12.

(a) $R = \sqrt{29}$

Accept 5.4 or 5.38, 5.39, 5.385....

B1

$$R \sin \alpha = 5 \text{ or } R \cos \alpha = 2 \text{ or } \tan \alpha = \frac{5}{2}$$

M1

$$\alpha = 68.2^\circ$$

Condone $\alpha = 68.20^\circ$

A1

3

- (b) (i) (maximum value $= \sqrt{29}$
ft on R

B1ft

1

(ii) $\sin(x + \alpha) = 1$

Or $x + \alpha = 90, x + \alpha = \frac{\pi}{2}$

M1

$x = 21.8^\circ$ only
No ISW

A1

[6]

Q13.

(a) (i) $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$

Condone numerator as $\tan x + \tan x$

B1

$$2 \tan x + \tan x (1 - \tan^2 x) = 0$$

Multiplying throughout by their denominator

M1

$$\tan x = 0$$

$$\text{or } (2 + 1 - \tan^2 x) = 0 \Rightarrow \tan^2 x = 3$$

AG Must show $\tan x = 0$ and $\tan^2 x = 3$

A1

3

Alternative

$$\tan 2x = \frac{\sin 2x}{\cos 2x} = \frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x}$$

$$\frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x} + \frac{\sin x}{\cos x} = 0$$

(B1)

$$2 \sin x \cos^2 x + \sin x (\cos^2 x - \sin^2 x) = 0$$

$$\sin x (2 \cos^2 x + \cos^2 x - \sin^2 x) = 0$$

(M1)

$$\left. \begin{array}{l} \Rightarrow \sin x = 0 \\ \Rightarrow \tan x = 0 \end{array} \right\} \text{ and } \left. \begin{array}{l} 3 \cos^2 x = \sin^2 x \\ \tan^2 x = 3 \end{array} \right\}$$

(A1)

(3)

(ii) $x = 60$ **AND** $x = 120$

*Condone extra answers outside interval
eg 0 and 180*

B1

1

(b) (i) $2 \sin x \cos x = \cos x f(x)$

Where $f(x) = \cos^2 x - \sin^2 x$

or $2 \cos^2 x - 1$ or $1 - 2 \sin^2 x$

M1

$$2 \sin x \cos x = \cos x (1 - 2 \sin^2 x)$$

A1

$(\cos x \neq 0) \quad 2 \sin x = 1 - 2 \sin^2 x$

$$2 \sin^2 x + 2 \sin x - 1 = 0$$

AG

A1

3

(ii) $\sin x = \frac{-2 \pm \sqrt{4 - 4 \times 2 \times (-1)}}{2 \times 2}$

Correct use of quadratic formula or completing the square or correct factors

M1

$$\sin x = \frac{-2 \pm 2\sqrt{3}}{4}$$

$\sqrt{12}$ must be simplified and must have $\pm$.

A1

$$\left. \begin{array}{l} \sin x = \frac{-1 - \sqrt{3}}{2} \text{ has no solution} \\ \sin x = \frac{\sqrt{3} - 1}{2} \end{array} \right\}$$

Reject one solution and state correct solution.

E1

3

[10]

Q14.

(a) (i) $\left(\frac{dx}{d\theta}\right) = -6 \sin 2\theta, \quad \left(\frac{dy}{d\theta}\right) = -2 \sin \theta$

$\left(\frac{dx}{d\theta}\right) = p \sin \theta \text{ or } r \sin \theta \cos \theta$
 $\left(\frac{dy}{d\theta}\right) = q \sin \theta$

M1

Both correct.

A1

$$\frac{dy}{dx} = \frac{-2 \sin \theta}{-6 \sin 2\theta}$$

$$\frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

Use chain rule ; condone one slip

M1

$$= \frac{2 \sin \theta}{6 \times 2 \sin \theta \cos \theta} = \frac{1}{6 \cos \theta}$$

$k = 6$ must come from correct working seen **AG**

A1

4

(ii) $\theta = \frac{\pi}{3} \quad m_T = \frac{1}{3}$

ft on $k \quad \left(\frac{1}{k \times \frac{1}{2}} \right)$

k need not be numerical

B1ft

$$m_N = -3$$

ft on m_T

B1ft

$$(x, y) = \left(-\frac{3}{2}, 1 \right)$$

B1

Normal $y - 1 = -3 \left(x + \frac{3}{2} \right)$

CAO; any correct form, ISW.

$$2y + 6x + 7 = 0$$

B1

4

(b) $\sin^2 x = \frac{1}{2} (1 - \cos 2x)$

$p + q \cos 2x$; Allow different letters for x or mixture eg θ even for A1 and the following A1ft

M1

A1

$$\int p \, dx = px \quad \int q \cos 2x = \frac{1}{2} q \sin 2x$$

Both integrals correct;

ft on p and q

A1ft

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx = \left[\frac{x}{2} - \frac{1}{4} \sin 2x \right]$$

$$= \left(\frac{\pi}{8} - \frac{1}{4} \right) - \left(-\frac{\pi}{8} - \left(-\frac{1}{4} \right) \right)$$

Correct use of limits;

$$F\left(\frac{\pi}{4}\right) - F\left(-\frac{\pi}{4}\right) \text{ or } 2F\left(\frac{\pi}{4}\right)$$

$$F(x) = px + r \sin 2x \text{ and } \sin \frac{\pi}{2}, \sin\left(-\frac{\pi}{2}\right)$$

must be evaluated correctly for m1

m1

$$= \frac{\pi}{4} - \frac{1}{2}$$

CSO OE ISW

A1

5

Alternative

$$\int \sin^2 x dx = -\sin x \cos x - \int -\cos x \cos x dx$$

Use parts; condone sign slips

(M1)

$$= -\sin x \cos x + \int 1 - \sin^2 x dx$$

Use $\cos^2 x = 1 - \sin^2 x$

(m1)

$$2 \int \sin^2 x dx = -\sin x \cos x + x$$

(A1)

$$2 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx = G\left(\frac{\pi}{4}\right) - G\left(-\frac{\pi}{4}\right)$$

Correct use of limits

(m1)

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx = \frac{\pi}{4} - \frac{1}{2}$$

(A1)

(5)

[13]

Q15.

(a) $R = \sqrt{10}$

Accept $R = 3.16$ or better

B1

$\tan \alpha = 3$

OE

M1

$\alpha = 1.249$ ignore extra out of range

*AWRT 1.25 SC $\alpha = 0.322$ B1
radians only*

A1

3

(b) (i) minimum value = $-\sqrt{10}$

F on R

B1F

1

(ii) $\cos(x - \alpha) = -1$

M1

$x = 4.391$

*AWRT 4.39
 51.56° or ... 57° or better*

A1F

2

(c) $\cos(x - \alpha) = \frac{2}{\sqrt{10}}$

M1

$x - \alpha = \pm 0.886 \quad 5.397$
 ignore extra out of range

*Two values, accept 2dp and condone 5.4
 condone use of degrees*

A1

$x = 0.36296.. \quad 2.13512..$

F on $x - \alpha$, either value. AWRT

A1F

$x = 0.363 \quad 2.135$

CSO

3dp or better

A1

Alternative

$10 \sin^2 x - 12 \sin x + 3 = 0$

*Or equivalent quadratic using $\cos x$
 (ie $\sin^2 x + \cos^2 x = 1$ used)*

M1

$\sin x =$ two numerical answers

Or equivalent using $\cos x$

A1F

$-1 \leq \text{ans} \leq 1$

$x =$ one correct answer

A1F

$x = 0.363 \quad 2.135$

CSO

3 dp or better

A1

4

[10]

Q16.

(a) (i) $\sin 2\theta = 2 \sin \theta \cos \theta$

B1

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

OE condone use of x etc, but variable must be consistent

B1

2

(ii) $\sin \theta = \frac{4}{5} \Rightarrow \sin 2\theta = 2 \times \frac{4}{5} \times \frac{3}{5} = \frac{24}{25}$

AG

Use of 106.26° B0

B1

or

$$2 \times \sin\left(\cos^{-1}\frac{3}{5}\right) \times \frac{3}{5}$$

$$\cos 2\theta = \frac{9}{25} - \frac{16}{25} = -\frac{7}{25}$$

-0.28

B1

2

(b) (i) $\frac{dx}{d\theta} = 8 \cos 2\theta$, $\frac{dy}{d\theta} = -8 \sin 2\theta$

Attempt both derivatives. ie p cos 2θ

M1

Both correct. q sin 2θ

A1

$$\frac{dy}{dx} = -\frac{4 \sin 2\theta}{3 \cos 2\theta} \quad \text{ISW}$$

CSO OE

A1

3

(ii) $P\left(\frac{72}{25}, -\frac{28}{25}\right)$

(2.88, -1.12)

B1F

$$\text{Gradient} = -\frac{4}{3} \times -\frac{24}{7}$$

$$\text{Their } \frac{q \sin 2\theta}{p \cos 2\theta} \quad \text{or} \quad \frac{p \cos 2\theta}{q \sin 2\theta}$$

must be working with rational numbers

M1

$$\text{Tangent } y + \frac{28}{25} = \frac{32}{7} \left(x - \frac{72}{25} \right) \quad \text{ISW}$$

Any correct form.

$$7y = 32x - 100$$

Fractions in simplest form

Equation required

A1

3

[10]

Q17.

(a) (i) $\cos 2x = 1 - 2 \sin^2 x$
ACF in terms of sin (PI later)

B1

$$3(1 - 2 \sin^2 x) + 2 \sin x + 1 = 0$$

Substitute candidate's cos 2x in terms of sin x (at least 2 terms)

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M1

$$-6 \sin^2 x + 2 \sin x + 4 = 0$$

$$3 \sin^2 x - \sin x - 2 = 0$$

AG

A1

3

(ii) $(3 \sin x + 2)(\sin x - 1) = 0$
Factorise correctly or use formula correctly

M1

$$\sin x = -\frac{2}{3} \quad \sin x = 1$$

Both; condone -0.67 or -0.66 or better

A1

2

(b) (i) $R = \sqrt{13}$
Accept 3.6 or better

B1

$$\tan \alpha = \frac{2}{3} \quad \alpha = 33.7$$

OE; accept $\alpha = 33.69(0)$

M1A1

3

(ii) $2x - \alpha = \cos^{-1}\left(\frac{-1}{R}\right)$

Candidate's R. Or $\cos(2x - \alpha) = \frac{-1}{R}$

M1

$$2x - \alpha = 106.1^\circ, 253.9^\circ$$

$$x = 69.9^\circ, 143.8^\circ$$

One correct answer

A1

Both correct, no extras in range

A1

3

[11]

Q18.

(a) $R = \sqrt{10}$

Accept $R = 3.16$ or better.

B1

$$\tan \alpha = 3$$

Oe (Can be implied by 71.57° seen)

M1

$$\alpha = 1.25$$

A0 if extra answers within given range

SC 1 $\tan \alpha = \frac{1}{3} \quad \alpha = 0.32$

A1

3

(b) (i) min value = $-\sqrt{10}$ (or $\geq \sqrt{-10}$)
ft or R

B1F

1

(ii) $\sin(x - \alpha) = -1$
 $x = 5.96$

or $\sin^{-1} \frac{3\pi}{2}$

M1

Ft on their α (to 2 dp) $+\frac{3\pi}{2}$

A1F

2

[6]

Q19.

(a) $\sin 2x = 2 \sin x \cos x$

OE, eg $\sin x \cos x + \sin x \cos x$ etc

B1

$\cos x = 0$ $x = 90, 270$
Both required

B1

1

(b) $10 \sin x + 3 = 0$

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M1

$x = 197.5$ 342.5

CAO

if extra values in given range, max $\frac{1}{2}$

A1A1

4

(c) $\cos 2x = \cos^2 x - \sin^2 x$

$\cos 2x$ in any correct form

B1

$2 \sin x \cos x + 1 - 2 \sin^2 x = 1 + \sin x$

$\sin 2x$ expanded and $\cos 2x$ in terms of

sin x used

M1A1

$$2 \sin x (\cos x - \sin x) = \sin x$$

$$2 (\cos x - \sin x) = 1$$

CSO; need to see sin x taken out as factor or cancelled

A1

4

[9]

Q20.

(a) (i) $\cos 2x = 2 \cos^2 x - 1$

Seen in question, in consistent variable

B1

$$3(2 \cos^2 x - 1) + 7 \cos x + 5$$

Substitute candidate's cos 2x in terms of cos x

M1

$$6 \cos^2 x + 7 \cos x + 2 (= 0)$$

A1

3

(ii) $(2 \cos x + 1)(3 \cos x + 2)$

Attempt factors; formula ('a' and 'c' correct; allow one slip)

M1

$$\cos x = -\frac{1}{2} \quad \cos x = -\frac{2}{3}$$

Accept -0.5, -0.67

$$x = \cos^{-1}\left(-\frac{1}{2}\right); \cos^{-1}\left(-\frac{2}{3}\right)$$

A1

2

(b) (i) $R = \sqrt{58}$

Accept 7.6 or better

B1

$$\alpha = \sin^{-1}\left(\frac{3}{\text{their } R}\right)$$

$$\text{OE } \alpha = \sin^{-1}\left(\frac{3}{7}\right)$$

M1

$$= 23.2^\circ$$

AWRT 23.2° (23.1985...)

A1

3

(ii) $\alpha + \theta = \sin^{-1}\left(\frac{4}{\text{their } R}\right)$

Candidate's R , α

M1

$$\theta = 8.5^\circ$$

F on α , AWRT, condone 8.6

A1F

$$\theta = 125.1^\circ$$

Two solutions only, but ignore out of range

A1

3

(c) (i) $h^2 = 1 + (2\sqrt{2})^2$

Pythagoras with h or $\sec x$

M1

$$h = 3 \Rightarrow \cos \beta = \frac{1}{3}$$

AG

A1

2

(ii) $\sin 2\beta = 2 \sin \beta \cos \beta$

m1

$$\sin 2\beta = \frac{4}{9}\sqrt{2}$$

CSO; accept $p = \frac{4}{9}$ (not 0.444...)

A1

2

[15]

Q21.

(a) (i) $R = 10$
 $R = 10$

B1

$\tan \alpha = \frac{8}{6}, \quad \alpha = 53.1$

For α ; ft incorrect R

B1F

2

(ii) $\sin (2x + 53.1) = 0.7$

M1

$2x + 53.1 = 44.4$

one correct answer ; ft α and R

A1F

135.6 or 135.7 , 404.4 , 495.6 or 495.7

3 other correct answers – ignore extras

A1

$x = 41.2$ or 41.3 , 175.6 or 175.7 ,
 221.2 or 221.3 , 355.6 or 355.7

four solutions

CAO (with decimal place discrepancies)

Answers only: 0/4

A1

4

(b) (i) $\sin 2x = 2 \sin x \cos x$

B1

$\cos 2x = \cos^2 x - \sin^2 x$

identities for $\sin 2x$ and $\cos 2x$ in any correct form

B1

$$\frac{\sin 2x}{1 - \cos 2x} = \frac{2 \sin x \cos x}{1 - (1 - 2 \sin^2 x)} =$$

use of candidate's double angle formulae

M1

$$\frac{2 \sin x \cos x}{2 \sin^2 x} = \frac{\cos x}{\sin x} = \frac{1}{\tan x}$$

AG, CSO

A1

4

(ii) $\frac{1}{\tan x} = \tan x \quad \tan x = \pm 1$
*(see * below)*

M1A1

$x = 45,$

B1

135, 225, 315

if answers given without working, B1 max

if $\frac{1}{\tan x} = \tan x$ seen and followed by correct answers without working 4 out of 4

A1

4

** If hence ignored, so working in sines and cosines, must simplify as far as:*

$\cos^2 x = \sin^2 x \quad \text{or} \quad \cos^2 x = \frac{1}{2} \quad \text{or} \quad \sin^2 x = \frac{1}{2} \quad \text{for M1}$			
$\cos 2x = 0 \quad \text{or} \quad \cos x = \pm \frac{1}{\sqrt{2}} \quad \text{or} \quad \sin x = \pm \frac{1}{\sqrt{2}} \quad \text{for A1}$			

[14]

Q22.

(a) $\sin(x + 2x) = \sin x \cos 2x + \cos x \sin 2x$

M1

$$= \sin x(1 - 2 \sin^2 x) + \cos x(2 \sin x \cos x)$$

double angles; ACF ISW condone missing x

B1B1

$$= \sin x(1 - 2 \sin^2 x) + 2 \sin x(1 - \sin^2 x)$$

all in sin x, correct expression

A1

$$= 3 \sin x - 2 \sin^3 x - 2 \sin^3 x$$

$$= 3 \sin x - 4 \sin^3 x$$

CSO AG

A1

5

(b) $\sin^3 x = a \sin x + b \sin 3x$

attempt to solve for $\sin^3 x$ where $a \neq 0$ and $b \neq 0$

M1

$$\int \sin^3 x \, dx = -a \cos x - \frac{b}{3} \cos 3x$$

*either integral correct
F on a, b*

A1F

$$\int \sin^3 x \, dx = \frac{1}{4} \left(-3 \cos x + \frac{1}{3} \cos 3x \right) (+C)$$

CAO

alternative method by parts (see end of mark scheme)

A1

ALTERNATIVE METHOD 1

$$\int \sin^3 x \, dx = \int \sin^2 x \sin x \, dx$$

identify parts and attempt to integrate

M1

$$= -\sin^2 x \cos x - \int -2 \cos x \sin x \cos x \, dx$$

$$= -\sin^2 x \cos x - \frac{2}{3} \cos^3 x \quad (+C)$$

A2

ALTERNATIVE METHOD 2

$$\int \sin^3 x dx = \int \sin^2 x d(-\cos x)$$

condone sign error

M1

$$= \int -(1 - \cos^2 x) d(\cos x)$$

$$= -\cos x + \frac{1}{3} \cos^3 x (+C)$$

A2

ALTERNATIVE METHOD 3

$$\int \sin x \sin^2 x dx$$

$$\int \sin x (1 - \cos^2 x) dx$$

this form and attempt to integrate

M1

$$= -\cos x + \frac{1}{3} \cos^3 x (+C)$$

A2

3

[8]

Q23.

(a) (i) $\cos \alpha = \frac{3}{5}$

ACF

B1

1

(ii) $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$

M1

$$= \frac{3}{5} \cos \beta + \frac{4}{5} \sin \beta$$

ACF

A1

2

(iii) $\sin \beta = \frac{12}{13}$

B1

$$\cos(\alpha - \beta) = \frac{63}{65}$$

$$\frac{63}{65} \quad \text{NMS B1 B1}$$

B1

2

(b) (i) $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$

M1

$$2 \tan x = 1 - \tan^2 x$$

$$\tan^2 x + 2 \tan x - 1 = 0$$

CSO AG

A1

2

(ii) $\tan x = \frac{-2 \pm \sqrt{4 + 4}}{2}$

*must solve quadratic equation by formula or by completing the square
condone one slip*

M1

$$= -1 \pm \sqrt{2}$$

$$\pm \sqrt{2} \text{ required}$$

A1

$$2x = 45^\circ \Rightarrow x = 22\frac{1}{2}^\circ \text{ is acute}$$

$$\Rightarrow \tan 22\frac{1}{2}^\circ = \sqrt{2} - 1$$

explain selection of positive root

E1

3

[10]

Q24.

(a) $\cos 2x = 1 - 2\sin^2 x$

B1

1

(b) (i) $3\sin x - \cos 2x = 3\sin x - (1 - 2\sin^2 x)$

Candidate's $\cos 2x$ or $\sin^2 x$

M1

$$= 3\sin x - 1 + 2\sin^2 x$$

AG

A1

2

(ii) $2\sin^2 x + 3\sin x - 2 = 0$

Soluble quadratic form

M1

$$(2\sin x - 1)(\sin x + 2) = 0$$

Attempt to solve (allow one error in formula, allow sign errors)

M1

$$\sin x = \frac{1}{2} \quad x = 30 \quad x = 150$$

M1

*$\sin^{-1}$ and two solutions ($0^\circ < x < 360^\circ$)
A0 if radians*

A1

4

Allow misread for $2\sin^2 x + 3\sin x - 1 = 0$

Soluble quadratic form

(M1)

$$\sin x = \frac{-3 \pm \sqrt{17}}{4}$$

Use of formula (allow one error)

(M1)

$$x = 16.3^\circ, 163.7^\circ$$

Max 3/4

(A1)

(c) $\int \frac{1}{2}(1 - \cos 2x) = \frac{x}{2} - \frac{\sin 2x}{4} (+c)$

*M1 – solve integral, must have 2 terms
for $\sin^2 x$ from (a)*

M1A1

2

[9]

Q25.

(a) $\tan(x+x) = \frac{\tan x + \tan x}{1 - \tan x \tan x} \left(= \frac{2 \tan x}{1 - \tan^2 x} \right)$

A = B = x used

M1
A1

2

(b) $2 - 2 \tan x - \frac{2 \tan x(1 - \tan^2 x)}{2 \tan x}$

Substitute from (a)

M1

$2 - 2 \tan x - (1 - \tan x)(1 + \tan x)$

Simplification $2 - 2 \tan x - (1 - \tan^2 x)$

M1

$(1 - \tan x)(2 - (1 + \tan x))$

$2 - 2 \tan x - 1 + \tan^2 x$

M1

$(1 - \tan x)^2$

AG (convincingly obtained)

$= (\tan x - 1)^2 = (1 - \tan x)^2$

Any equivalent method

A1

4

[6]

Q26.

(a) $R = 5$

B1

$$\tan \alpha = \frac{3}{4} \text{ (OE)} \quad \alpha = 36.9^\circ \text{ (ISW 216.9)}$$

$$\text{SC1 } \tan \alpha = \frac{4}{3}, \alpha = 53.1^\circ$$

R, α , PI in (b)

M1A1

3

(b) $\cos(x - \alpha) = \frac{2}{R}$

M1

$$x - \alpha = 66.4^\circ$$

A1

$$x = 103.3^\circ$$

A1F

$$x = 330.4^\circ$$

*accept 330.5°, -1 each extra
ft on acute α*

A1F

4

(c) minimum value = -5

ft on R

B1F

$$\cos(x - 36.9) = -1$$

SC $\cos(x + 36.9)$ treat as miscopy

M1

$$x = 216.9^\circ$$

*216.9 or better accept graphics calculator
solution to this accuracy*

SC Find max:

max = 5 at $(x + 36.9) + \text{stated } 1/3$

Max 8/10 for work in radians

A1

3

[10]

Q27.

(a) $R = \sqrt{13}$ Or 3.6

B1

1

(b) $\frac{\sin \alpha}{\cos \alpha} = \tan \alpha = \frac{2}{3}$ $\alpha \approx 33.7$

Allow M1 for $\tan \alpha = \frac{-2}{3}$ or $\pm \frac{3}{2}$
 AG convincingly obtained

M1A1

2

(c) maximum value = $\sqrt{13}$

B1F

$\cos(\theta + 33.7) = 1$ ($\theta = -33.7$)

M1

$\theta = 326.3$

AWRT 326

A1

3

[6]

Q28.

(a) $\cos 2x = 2 \cos^2 x - 1$

B1B1

2

(b) $\cos^2 x = \frac{1}{2} (\cos 2x + 1)$

Attempt to express $\cos^2 x$ in terms of $\cos 2x$

M1

$$\frac{1}{2} \int_0^{\frac{\pi}{2}} \cos 2x + 1 \, dx = \left[\frac{1}{4} \sin 2x + \frac{x}{2} \right]_0^{\frac{\pi}{2}}$$

A1

A1

$$= \frac{\pi}{4}$$

Use limits. Ft on integer a.

M1A1F

5

[7]

Q29.

(a) (i) $\sin 2x = 2 \sin x \cos x$

B1

1

(ii) $\cos 2x = 2 \cos^2 x - 1$

B1

1

(b) $\sin 2x - \tan x = 2 \sin x \cos x - \frac{\sin x}{\cos x}$

Use of their $\cos 2x$ or $\sin 2x$

M1

$$= \sin x \left(2 \cos x - \frac{1}{\cos x} \right)$$

Use of $\tan x = \frac{\sin x}{\cos x}$ and the other double angle identity

M1

$$= \sin x \left(\frac{2 \cos^2 x - 1}{\cos x} \right) = \tan x \cos 2x$$

AG convincingly obtained

A1

3

(c) $\tan x \cos 2x = 0 \quad x = 180$

Ignore $x = 0$, $x = 360^\circ$ & any others outside range

B1

$$\cos 2x = 0 \text{ or } \cos^2 x = \frac{1}{2} \left(\text{or } \sin^2 x = \frac{1}{2} \right)$$

M1

$$x = 45$$

A1

$$x = 135, 225, 315$$

CAO max 3/4 for answers in radians

B1

4

[9]

Q30.

(a) $R = \sqrt{5}$ (or $\sqrt{1+2^2}$ or 2.23 or 2.24)

B1

$$\frac{\sin \alpha}{\cos \alpha} = \frac{1}{2} \quad \alpha = 26.6^\circ$$

CAO
SC 63.4° 1/2

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M1A1

3

(b) $\sin (x + 26.6) = \frac{1}{\sqrt{5}}$

M1

$$x = \sin^{-1} \left(\frac{1}{\sqrt{5}} \right) - 26.6$$

B1 $x = 0^\circ$

m1

$$x = 0^\circ \quad x = 126.8^\circ (126.9^\circ, \text{ or } 127 \text{ or } 126^\circ \text{ with working})$$

lose if extra solution in range

SC calculator trace: 126.9 full marks

SC ft from 63.4°

B1

A1

4

[7]

Q31.

(a) $\sin 2x = 2 \sin x \cos x$

B1

1

(b) (i) $A = B = x$

M1

$$\cos 2x = \cos^2 x - \sin^2 x$$

A1

2

(ii) $\cos (2x + x) = \cos 2x \cos x - \sin 2x \sin x$

M1

$$= (\cos^2 x - \sin^2 x) \cos x - 2 \sin^2 x \cos x$$

ft (a) and (b)(i)

A1ft

$$= \cos^3 x - 3 (1 - \cos^2 x) \cos x$$

Eliminate all sines

M1

$$= 4 \cos^3 x - 3 \cos x$$

AG; convincingly obtained

A1

4

(c) $\cos^3 x = \frac{1}{4} (\cos 3x + 3 \cos x)$

Ignore middle sign

M1

$$\int_0^{\frac{\pi}{2}} \cos^3 x dx = \frac{1}{4} \left[\frac{1}{3} \sin 3x + 3 \sin x \right]_0^{\frac{\pi}{2}}$$

1 for integrating each term

A1A1

Use of limits

M1

$$\frac{1}{4} \left(-\frac{1}{3} + 3 \right) = \frac{2}{3}$$

AG

A1

5

[12]

Q32.

(a) $0 = V \sin \alpha t - gt^2$

Equation for height equal to zero

B1

$$t = 0 \text{ or } t = \frac{2V \sin \alpha}{g}$$

Finding t

Correct time of flight

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M1 A1

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$$\text{Range} = V \cos \alpha \times \frac{2V \sin \alpha}{g}$$

Expression for horizontal displacement

Substituting the time of flight

M1 M1

$$= \frac{V^2 \sin 2\alpha}{g}$$

Correct final answer from correct working

A1

6

(b) Because the maximum value of $\sin 2\alpha$ is 1.

Reason that uses maximum of sin being 1

B1

$$\alpha = 45^\circ$$

$$45^\circ$$

B1

2

(c) (i) $\frac{V^2}{g} = 35$
Substituting 35

M1

$$V = \sqrt{35g} = 18.5 \text{ m s}^{-1}$$

cao

A1

2

(ii) $R = \frac{18.5^2 \sin 60^\circ}{g} = 30 \text{ m}$
Substituting $\alpha = 30^\circ$
30 m

M1 A1

2

(or $R = 35 \sin 60^\circ = 30$)

[12]